

Kiran M. Jayasurya

Scientist, Space Astronomy Group, U R Rao Satellite Centre (URSC), ISRO, Bengaluru · PhD researcher, Raman Research Institute

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PROFILE

I am a Scientist in the Space Astronomy Group at the U R Rao Satellite Centre (URSC), ISRO, Bengaluru, and am pursuing a PhD at the Raman Research Institute (RRI). My research interests include X-ray binaries, pulsars, ultraluminous X-ray sources, X-ray spectro-polarimetry, and X-ray instrumentation.

RESEARCH INTERESTS

X-ray binaries, pulsars, and ultraluminous X-ray sources; tidal disruption events around supermassive black holes; gamma-ray bursts and multi-messenger astronomy; and instrumentation for broadband X-ray spectrometers and photoelectric polarimeters.

PROFESSIONAL APPOINTMENT

Scientist — U R Rao Satellite Centre (URSC), ISRO | January 2021–Present

Space Astronomy Group. Involved in the development of space-based astronomy payloads for future ISRO missions.

RESEARCH EXPERIENCE

Indian Institute of Space Science and Technology, Thiruvananthapuram | 2018–2020

Postgraduate researcher — tidal disruption events around supermassive black holes.

Indian Institute of Space Science and Technology, Thiruvananthapuram | 2017–2020

Student researcher — gamma-ray burst afterglow modelling.

Physical Research Laboratory | June 2018

Student researcher — accretion disks around compact objects.

EDUCATION

Raman Research Institute, Bengaluru | 2025–Present

PhD in Astronomy and Astrophysics.

Indian Institute of Space Science and Technology, Thiruvananthapuram | 2018–2020

Master of Science in Astronomy and Astrophysics.

Indian Institute of Space Science and Technology, Thiruvananthapuram | 2015–2018

BTech in Engineering Physics.

TECHNICAL SKILLS

Programming and scientific computing: Python, MATLAB, C++, LaTeX. · Simulation and detector tools: Geant4, Garfield++, COMSOL. · Astronomy software: HEASoft, CIAO, XMM-Newton SAS, HENDRICS.

SELECTED PEER-REVIEWED PUBLICATIONS

- Chatterjee, R., Vadodariya, K., Vatedka, R., Agrawal, V. K., Tyagi, A., Jayasurya, K. M., et al. (2025). "XSPECT on-board XPoSat: Calibration and First Results." arXiv, arXiv:2506.09918. doi:10.48550/arXiv.2506.09918.
- Vatedka, R., Tyagi, A., Vadodariya, K., Agrawal, V. K., Chatterjee, R., Chellammal, R. M., Jayasurya, K. M., et al. (2025). "X-ray spectroscopy and timing experiment on XPoSat: instrument configuration and science prospects." Journal of Astronomical Telescopes, Instruments, and Systems, 11(3), 035001. doi:10.1117/1.JATIS.11.3.035001.
- Majumder, S., Kushwaha, A., Singh, S., Jayasurya, K. M., Das, S., & Nandi, A. (2025). "Probing the accretion geometry of black hole X-ray binaries: A multi-mission spectro-polarimetric and timing study." arXiv, arXiv:2506.03774.
- V., R., Tyagi, A., Vadodariya, K., Agrawal, V. K., Chatterjee, R., et al. (2025). "X-ray spectroscopy and timing (XSPECT) experiment on XPoSat — instrument configuration and science prospects." arXiv, arXiv:2505.20061.
- Majumder, S., Chatterjee, R., Jayasurya, K. M., Das, S., & Nandi, A. (2024). "First Detection of X-Ray Polarization in Galactic Ultraluminous X-Ray Pulsar Swift J0243.6+6124 with IXPE." The Astrophysical Journal Letters, 971(1), L21. doi:10.3847/2041-8213/ad67e5.
- Jayasurya, K. M., Agrawal, V. K., & Chatterjee, R. (2023). "Detection of significant X-ray polarization from transient NS-LMXB XTE J1701-462 with IXPE and its implication on the coronal geometry." Monthly Notices of the Royal Astronomical Society, 525(3), 4657–4662. doi:10.1093/mnras/stad2601.

7. Kushwaha, A., Jayasurya, K. M., & Nandi, A. (2023). "Spectro-polarimetry of 4U 1957+115 with IXPE: Effects of spin and returning radiation on polarised emission of black hole in thermal state." arXiv, arXiv:2310.02222.
8. Kushwaha, A., Jayasurya, K. M., Agrawal, V. K., & Nandi, A. (2023). "IXPE and NICER view of black hole X-ray binary 4U 1630-47: First significant detection of polarized emission in thermal state." Monthly Notices of the Royal Astronomical Society: Letters, 524(1), L15–L20. doi:10.1093/mnrasl/slad070.
9. Chatterjee, R., Agrawal, V. K., Jayasurya, K. M., & Katoch, T. (2023). "Spectro-polarimetric view of bright atoll source GX 9+9 using IXPE and AstroSat." Monthly Notices of the Royal Astronomical Society: Letters, 521(1), L74–L78. doi:10.1093/mnrasl/slad026.

OTHER PUBLICATIONS AND PROCEEDINGS

- Jayasurya, K. M., Vadodariya, K., & Radhakrishna, V. (2024). "Multifaceted Simulations to design a TPC Polarimeter for Soft to Medium X-rays." 42nd meeting of the Astronomical Society of India, 42, P11.
- Surabhi, C., Resmi, L., & Jayasurya, K. M. (2024). "Exploring Gamma-Ray Burst Radio Afterglow Population." 42nd meeting of the Astronomical Society of India, 42, P137.
- Majumder, S., Chatterjee, R., Jayasurya, K., Das, S., & Nandi, A. (2024). "First IXPE view of ULX pulsar Swift J0243.6+6124: Polarimetric results and detection of pulsation." 45th COSPAR Scientific Assembly, 45, 1586.
- Perley, D., Copperwheat, C., Guolo-Pereira, M., et al., including Jayasurya, K. M. (2020). "Follow-up observations of GSNST-20ca (AT2020rml) with the Liverpool Telescope." Transient Name Server AstroNote, 161, 1.

CONFERENCES AND WORKSHOPS

XSPECT-XPoSat Workshop (2025), URSC, Bengaluru; Astronomical Society of India (2025), NIT Rourkela; NICER-IXPE Workshop (2024), Washington, D.C.; National Space Science Symposium (2024), University of Goa, Panaji; Astronomical Society of India (2024), IISc Bengaluru; GROWTH Time Domain Astronomy Workshop (August 2020), California Institute of Technology.